

CorpayOne → NetSuite Integration

Solution Overview & Demo Output

1. Architecture Overview

This integration provides a **one-way sync from CorpayOne to NetSuite**. CorpayOne is read-only — all mapping configuration lives in NetSuite (via a SuiteScript Suitelet dashboard or the REST API). When invoices arrive from CorpayOne, they are automatically mapped to NetSuite Vendor Bills with the correct GL accounts, tax codes, bank accounts, and subsidiaries.

Data Flow: CorpayOne API → Middleware → NetSuite REST API

2. Key Design Decisions

Category-based GL Account Resolution

The mapping schema uses the CorpayOne expense **category** (e.g. "IT Equipment", "Cloud Services") as the primary key for GL account resolution, rather than internal account codes. This matches what users see in CorpayOne and simplifies admin setup.

Multi-level Fallback Chain

1. Category + Subsidiary
2. Category only
3. Account code + Subsidiary
4. Account code only
5. Default mapping (`is_default = 1`)
6. Static config (`NETSUITE_AP_ACCOUNT_ID` env var)

VAT/Tax Amount Passthrough

The **vat_amount** from each CorpayOne line item is always passed through to NetSuite as **taxAmount** on the expense line, preserving the original tax figure from the source invoice.

3. Database Schema (account_mappings)

```
CREATE TABLE IF NOT EXISTS account_mappings (  
  id INTEGER PRIMARY KEY AUTOINCREMENT,  
  corpayone_category TEXT NOT NULL,      -- Primary lookup key  
  corpayone_account_code TEXT,          -- Fallback lookup  
  netsuite_account_id TEXT NOT NULL,  
  netsuite_account_name TEXT,  
  subsidiary_id TEXT,  
  is_default INTEGER NOT NULL DEFAULT 0,  
  UNIQUE(corpayone_category, subsidiary_id)  
);
```

4. Mapping Configuration (NetSuite Dashboard)

Admins use the CorpayOne Configuration Suitelet in NetSuite to set up mappings:

CorpayOne Category	NetSuite GL Account	NS Account ID
IT Equipment	IT Equipment	201
Office Supplies	Office Supplies	202
Office Furniture	Office Furniture	203
Cloud Services	Cloud & Hosting Services	301
Default	Accounts Payable	400

VAT Rate	Country	NetSuite Tax Code
25%	DK	DK Moms 25%
19%	DE	DE Umsatzsteuer 19%
20%	GB	GB VAT 20%

5. Files Modified

src/database/mapping-schema.ts	Schema: category as primary key
src/database/mapping-db.ts	Multi-level fallback resolver
src/mapping/invoice-mapper.ts	Category-first GL resolution + VAT passthrough
netsuite-suitescript/ corpay_config_suitelet.js	Dashboard: category columns
demo/run-demo.ts	Demo: Net/VAT/Total per line

6. Demo Output: Vendor Bills with VAT Breakdown

Each vendor bill shows net amount, VAT, and total per expense line:

Invoice	Account	Net	VAT	Tax Code
TS-2025-0042	IT Equipment (201)	6,000	1,500	DK Moms 25%
	Office Supplies (202)	2,000	500	DK Moms 25%
	TOTAL	8,000	2,000	Grand: 10,000
CH-8834	Cloud & Hosting (301)	1,500	285	DE USt 19%
	Cloud & Hosting (301)	1,000	190	DE USt 19%
	TOTAL	2,500	475	Grand: 2,975
OS-2025-771	Office Furniture (203)	1,200	240	GB VAT 20%
	TOTAL	1,200	240	Grand: 1,440

7. Full Demo Output (Steps 1-7)

The integration demo runs 7 steps with mock data. Below is the captured terminal output:

```
CorpayOne → NetSuite Integration Demo
```

```
This demo simulates the full integration flow with mock data.  
No real API credentials needed.
```

```
Step 1: Configure Mapping in NetSuite (one-way: CorpayOne → NetSuite)  
■ An admin opens the CorpayOne Configuration Suitelet in NetSuite.  
■ Nothing is configured in CorpayOne – it is read-only.  
■ The admin maps CorpayOne expense categories to NetSuite GL accounts,  
■ CorpayOne VAT rates to NetSuite tax codes, and selects bank accounts.
```

```
Configuring subsidiary...  
✓ Subsidiary: Corpay Denmark ApS (NS ID: 1) → Default
```

```
Mapping CorpayOne categories → NetSuite GL accounts...  
■ These categories come from CorpayOne invoice line items automatically.
```

```
✓ "IT Equipment" → NS: IT Equipment (201)  
✓ "Office Supplies" → NS: Office Supplies (202)  
✓ "Office Furniture" → NS: Office Furniture (203)  
✓ "Cloud Services" → NS: Cloud & Hosting Services (301)  
✓ "Default" → NS: Accounts Payable (400) [DEFAULT]
```

```
Configuring tax code mappings...  
✓ 25% (DK Moms 25%) → NS: DK Moms 25%  
✓ 19% (DE Umsatzsteuer 19%) → NS: DE Umsatzsteuer 19%  
✓ 20% (GB VAT 20%) → NS: GB VAT 20%
```

```
Configuring bank accounts for payments...  
■ ✓ DKK → NS: CorpayOne Bank (DKK) (152) [DEFAULT]  
■ ✓ EUR → NS: CorpayOne Bank (EUR) (153)  
■ ✓ GBP → NS: CorpayOne Bank (GBP) (154)
```

ID	Number	Vendor	Status	Net	VAT	Total
inv-1001	TS-2025-0042	TechSupply ApS	approved	8000 DKK	2000 DKK	10000 DKK
inv-1002	CH-8834	CloudHost GmbH	approved	2500 EUR	475 EUR	2975 EUR
inv-1003	OS-2025-771	Office Solutions Ltd	paid	1200 GBP	240 GBP	1440 GBP
inv-1004	DRAFT-99	TechSupply ApS	draft	500 DKK	125 DKK	625 DKK

```
Fetches 4 invoices from CorpayOne  
Done
```

```
Step 3: Sync Vendors to NetSuite  
For each syncable invoice, ensure the vendor exists in NetSuite.
```

```
Looking up vendor "TechSupply ApS" (vendor-001)...  
SuiteQL: SELECT id FROM vendor WHERE externalid = 'corpay-vendor-vendor-001'  
→ Not found. Creating new vendor...  
✓ Created vendor "TechSupply ApS" → NS Internal ID: 5000  
Looking up vendor "CloudHost GmbH" (vendor-002)...  
SuiteQL: SELECT id FROM vendor WHERE externalid = 'corpay-vendor-vendor-002'  
→ Not found. Creating new vendor...  
✓ Created vendor "CloudHost GmbH" → NS Internal ID: 5001  
Looking up vendor "Office Solutions Ltd" (vendor-003)...  
SuiteQL: SELECT id FROM vendor WHERE externalid = 'corpay-vendor-vendor-003'  
→ Not found. Creating new vendor...  
✓ Created vendor "Office Solutions Ltd" → NS Internal ID: 5002  
Done
```

```

■ ✓ inv-1001 (TS-2025-0042) → NS Vendor Bill #5003
■   → Account: IT Equipment (201) | Net: 6000 | VAT: 1500 | Tax Code: DK Moms 25%
■   → Account: Office Supplies (202) | Net: 2000 | VAT: 500 | Tax Code: DK Moms 25%
■     ■ Total Net: 8000 | Total VAT: 2000 | Grand Total: 10000
■   → Subsidiary: Corpay Denmark ApS
■ ✓ inv-1002 (CH-8834) → NS Vendor Bill #5004
■   → Account: Cloud & Hosting Services (301) | Net: 1500 | VAT: 285 | Tax Code: DE Umsatzsteuer 19%
■   → Account: Cloud & Hosting Services (301) | Net: 1000 | VAT: 190 | Tax Code: DE Umsatzsteuer 19%
■     ■ Total Net: 2500 | Total VAT: 475 | Grand Total: 2975
■   → Subsidiary: Corpay Denmark ApS
■ ✓ inv-1003 (OS-2025-771) → NS Vendor Bill #5005
■   → Account: Office Furniture (203) | Net: 1200 | VAT: 240 | Tax Code: GB VAT 20%
■     ■ Total Net: 1200 | Total VAT: 240 | Grand Total: 1440
■   → Subsidiary: Corpay Denmark ApS
■ ■ inv-1004 (DRAFT-99) – status "draft" → Skipped

```

- Map each completed payment to a NetSuite Vendor Payment.

```

■ ✓ pay-2001 → NS Vendor Payment #5006
■   → Applied to Bill #5005 | Amount: 1440 GBP
■   → Bank Account: CorpayOne Bank Account (GBP) (154)
■   → Payment Date: 03/18/2025
■ ■ pay-2002 – status "pending" → Skipped (only completed payments sync)
■

```

- Step 6: Handle Real-Time Webhook Event
 - Simulating a webhook from CorpayOne: invoice.approved

```

■ Webhook payload: {
■   "id": "evt-9001",
■   "type": "invoice.approved",
■   "data": {
■     "invoice_id": "inv-1005"
■   },
■   "created_at": "2026-02-18T18:33:06.175Z"
■ }

```

```

■ Fetching invoice inv-1005 from CorpayOne...
■ ✓ Webhook processed: inv-1005 (TS-2025-0099) → NS Bill #5007
■   → Account: IT Equipment | Net: 4000 | VAT: 1000 | Tax Code: DK Moms 25%
■■ Done

```

```

■ NetSuite Vendors:
■ NS ID   Company Name           External ID
■
■ 5000    TechSupply ApS               corpay-vendor-vendor-001
■ 5001    CloudHost GmbH              corpay-vendor-vendor-002
■ 5002    Office Solutions Ltd        corpay-vendor-vendor-003
■

```

■	NS ID	Vendor ID	Tran ID	Total	External ID
■					
■	5003	5000	TS-2025-0042	8000	corpay-inv-1001
■	5004	5001	CH-8834	2500	corpay-inv-1002
■	5005	5002	OS-2025-771	1200	corpay-inv-1003
■	5007	5000	TS-2025-0099	4000	corpay-inv-1005
■					

NS ID	Vendor ID	Applied To Bill	Amount	External ID
5006	5002	5005	1440	corpay-pay-pay-2001

```

■ Account mappings:      5
■ Tax code mappings:    3
■ Bank accounts:        3
■ Subsidiaries:         1
■■ Done

```

[illegible]

In production, this same flow runs automatically via:

- Scheduled sync every 15 minutes (configurable)
- Real-time webhooks from CorpayOne
- NetSuite Suitelet dashboard for mapping configuration
- REST API at /api/config/* for programmatic access

8. Summary

The CorpayOne → NetSuite integration successfully:

- Reads invoices from CorpayOne (read-only, no write-back)
- Maps expense **categories** to NetSuite GL accounts using configurable rules
- Preserves **VAT amounts** from source invoices on each expense line
- Syncs vendor bills and payments to NetSuite with full subsidiary support
- Handles real-time webhooks for immediate invoice processing
- Provides a NetSuite Suitelet dashboard for admin configuration
- Exposes a REST API at `/api/config/*` for programmatic access

All changes committed and pushed to branch: `claude/corpayone-netsuite-integration-r0GiH`